

# Project Summaries

## 1. Friend's Project Summary – Decentralized Identity Verification Using Deep Learning Biometrics and Blockchain

This research project focuses on creating a smarter and safer way to verify a person's identity online. Traditional systems store user information in a centralized database, making them vulnerable to cyberattacks. The project combines Deep Learning and Blockchain to solve this problem. Deep Learning, using a Convolutional Neural Network (CNN), recognizes biometric features such as a face or fingerprint instead of relying only on passwords. This provides faster, more accurate, and more secure identity verification. The biometric data is converted into a SHA-256 hash, and only this encrypted hash is stored on the blockchain rather than the actual biometric data. This protects user privacy while ensuring the information cannot be secretly changed or deleted. During authentication, a new biometric sample is converted into a hash and compared with the blockchain record. If both hashes match, the user's identity is verified. Smart contracts automate identity registration, verification, and revocation without depending on a central authority. The project uses Python, TensorFlow, OpenCV, Solidity, Ethereum Ganache, Web3.py, and SHA-256 hashing. The prototype successfully demonstrates secure identity verification, privacy protection, and a permanent audit trail for cybersecurity and digital forensics. Future enhancements include multi-modal biometrics, public blockchain deployment, federated learning, and zero-knowledge proofs.

## 2. My Project Summary – Blockchain-Secured Deep Learning Framework for Cyber Forensic Evidence Management

This research project proposes a Blockchain-Secured Deep Learning Framework for Cyber Forensic Evidence Management. The main objective is to protect digital evidence from tampering while improving cyber threat detection during cybercrime investigations. Traditional forensic systems rely on centralized storage, making digital evidence vulnerable to unauthorized modification and single points of failure. To overcome these limitations, the proposed framework combines Blockchain and Deep Learning. Digital evidence such as system logs, network traffic, and access records is first collected and analyzed using Deep Learning models like CNN and LSTM. These models identify suspicious activities, abnormal user behavior, and potential cyberattacks with better accuracy than traditional rule-based methods. Once the evidence is collected, a unique SHA-256 hash is generated for each evidence file. Instead of storing the actual evidence on the blockchain, only its cryptographic hash is stored. This ensures that if anyone modifies the evidence later, the regenerated hash will not match the blockchain record, immediately revealing tampering. Blockchain also maintains a secure and transparent chain of custody, while smart contracts automate evidence registration, verification, and audit tracking. The implementation uses Python, TensorFlow, Ethereum, Ganache, Solidity, Web3.py, and SHA-256 hashing in a simulated environment. The results demonstrate improved evidence integrity, secure auditability, transparent investigations, and effective anomaly detection. Overall, the project shows how Blockchain and Artificial Intelligence can work together to build a secure, tamper-resistant, and intelligent cyber forensic evidence management system with strong potential for future real-world adoption.